

ian returns. According to her study, the classic cross-sectional contrarian strategy results could be overturned in the US stock market when controlling for January seasonality. In that sense, we also examine the impact of January seasonality on time series reversal (TSM and RSM) reporting in Section C of the Online Appendix. We find that the profitability of time series reversal is not centralised in January, but is shown in the other 11 months as well. This can be explained by the fact that our universe includes different types of futures contracts whereas the January effect is more commonly found in equity markets.

Since these contrarian strategies are performed using the volatility scaling method, low volatile instruments, e.g., bond futures, may dominate the portfolio. As shown in Table 1, the volatilities of bond futures are mostly below 10%, which are much lower than the volatilities of the three remaining asset classes. Therefore, we also apply contrarian strategies the same as in Figure 4 using the sample portfolio but excluding all bond futures. We note that time series reversal still holds after controlling for bond futures. These results are presented in Section D of the Online Appendix.

In summary, time series reversal is a pervasive but not economically significant phenomenon across the two strategies and time horizons. Normally, it occurs between the end of the first year and the end of the second year, after the trend-following signal is generated. The results differ from those in the existing literature, where the long-term reversals, cross-sectional or time series, usually last for much longer.¹³ Time series reversal vanishes after the end of the second year.

4. Linkage Between Time Series Momentum and Reversal

In this section, two approaches are adopted to investigate the linkage between time series momentum and reversal: the sub-portfolio sign analysis and the performance evaluation. These two analyses further track the decomposed trend-following

¹³For example, cross-sectional reversals require a long-term ranking period of 2–5 years in stock markets (De Bondt and Thaler, 1985) and 1.5–3 years in commodity markets (Bianchi et al., 2015). Bhojraj and Swaminathan (2006) find that the reversals of country indices occur from 2 to 3 years after the 1-year momentum effect. Moskowitz et al. (2012) document a statistically insignificant long-term (2–5 years) TSM reversal.